Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

Section A **2*5=10**

1)	Define software ethics. Mention any two principles of software development.	**2**	(CO1, L3)
2)	Explain the concept of a database transaction. What are its main properties?	**2**	(CO2, L3)
3)	What is DDL? Name any two commands used in DDL.	**2**	(CO3, L2)
4)	What is the difference between a primary key and a foreign key in database design?	**2**	(CO3, L3)
5)	List the stages involved in the software development life cycle. Briefly describe any two.	**2**	(CO1, L3)

Section B **5*4=20**

6)	What are the main components of a relational database management system (RDBMS)? OR What is normalization in DBMS and why is it important?	**5**	(CO2, L4)
7) a)	Explain the concept of normalization in database design and discuss its levels (1NF to 5NF). OR What are SQL queries? Name two types of SQL commands.	**5**	(CO3, L4)
	OR		
b)	Explain the concept of normalization in database design and discuss its levels (1NF to 5NF). OR What are SQL queries? Name two types of SQL commands.	**5**	(CO3, L4)

Section C **10*2=20**

8) a)	Design a simple software development project that adheres to ethical guidelines. Describe the project's purpose and the steps you would take to ensure it complies with ethical standards.	**10**	(CO1, L5)
	OR		
b)	Construct a database project using DDL and DML queries. Describe the entities involved, their relationships, and provide sample SQL commands for creating tables and inserting data.	**10**	(CO3, L4)

AI-assisted draft - teacher review and approval required before use

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create